

Weekly Progress Update

Recovery status, seed provenance, and the four-seed DistDNA validation plan

Reporting period: 5–13 August 2026 · Snapshot: 13 August, 13:56 SGT

This note updates the 5 August report. It records only what changed during the week, what the current data can support, and the next experiment needed to test DistDNA fairly. The full experimental protocol and previously reported formulas remain in the earlier report and are not repeated here.

2,552 successful artifacts of 2,600 planned	+24 failed slots recovered since 5 August	68 models complete across all 26 planned runs	48 slots still lacking a successful artifact
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What changed this week

- ✓ A failure-specific recovery pass was launched on the current A100 host and increased grid coverage from 97.2% to 98.2%.
- ✓ The strict all-run cohort increased from 59 to 68 models. Recovery is still active, so this is a timestamped interim count.
- ✓ Seed provenance was separated from artifact coverage: 1,956 successful slots still rely only on historical runs with no recorded generation seed.
- ✓ The repository now contains the formal RFFTrace/DistDNA implementation, exact MMD baseline, matched-sample evaluator, compact projection, and tests.
- ✓ A leakage-free four-seed pilot is specified so that DistDNA and cosine baselines consume exactly the same generated responses.

Recommended decision. First run a 10-model, one-setting, four-seed debugging pilot. Use it to validate the data split, implementation, and effect direction. Then start the 100-model extension at the same setting. Expand to all 12 stochastic settings only after the pilot is stable.

CURRENT JOB	Recovery was running at report build time. The active cell had moved to T=0.5, top-p=0.9, repeat 1 , using generation seed 42 .
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1. Recovery progress since 5 August

Measure	5 Aug frozen snapshot	13 Aug live snapshot	Change
Successful artifacts	2,528 / 2,600	2,552 / 2,600	+24
Artifact coverage	97.2%	98.2%	+1.0 pp
Slots without success	72	48	-24
Strict all-26-run cohort	59 / 100	68 / 100	+9 models
Explicit-seed successes	572	596	+24
Historical unknown-seed only	1,956	1,956	unchanged

Primary robustness grid — live recovery snapshot

Generated 2026-08-13 13:56:24+08:00; 100 models × 13 settings × 2 rounds

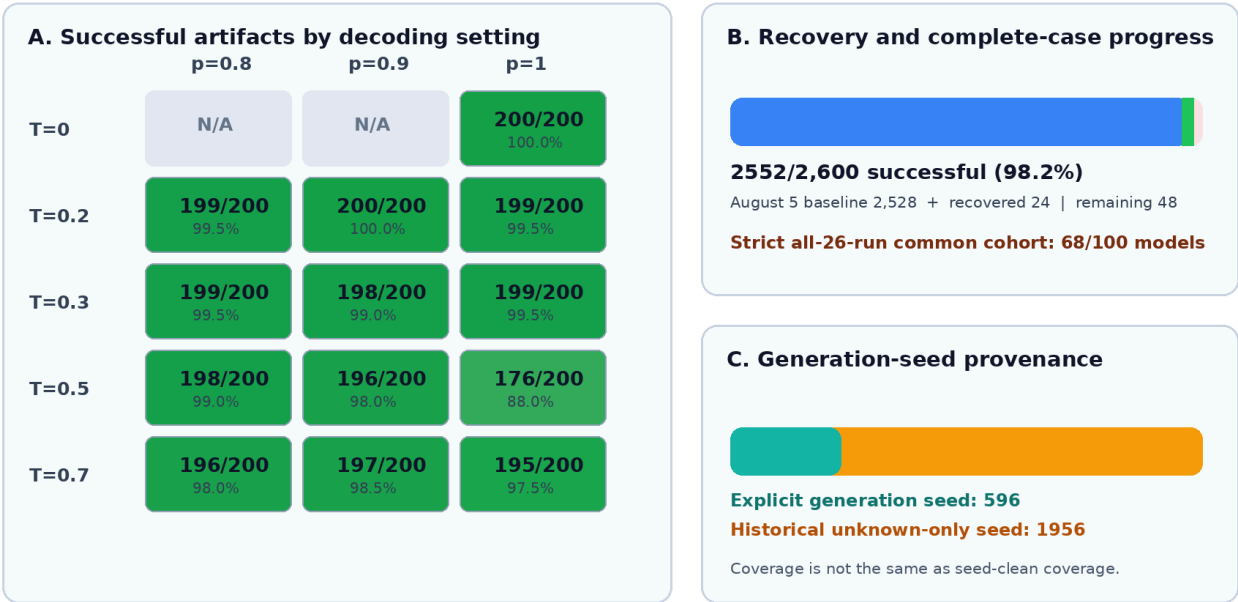


Figure 1. Live grid snapshot at 13:56 SGT on 13 August. Green recovery gains are real successful artifacts, but the seed-provenance panel shows why high coverage is not yet equivalent to a four-seed DistDNA dataset.

Interpretation. The recovery pass repairs the preregistered two-round grid. It should finish before a final primary-grid analysis is frozen. It does not, by itself, create four independent samples per model and setting.

Settings that remain most incomplete

The largest remaining gap is **T=0.5, top-p=1.0** with 176/200 successful artifacts. All other stochastic settings are between 195/200 and 200/200. Failures are distributed across models, which is why the strict common cohort remains lower than the overall coverage percentage.

2. Seed provenance: what is reusable

The response-generation seed and the DNA projection seed are different variables. This report concerns the response-generation seed. The projection seed remains fixed and does not create an independent response sample.

Artifact source	Recorded generation seed	Use in current analysis	Use in four-seed DistDNA
Stochastic T=0.2, 0.3, 0.7 backfills	r1: 42001 r2: 42002	Valid	Reusable as independent seeds
T=0.5 control and recovery	r1: 42 r2: 42	Valid for coverage	Replace for seed-clean study
Historical sweep	Not recorded	Valid for matched method comparisons on the same responses	Replace; independence cannot be established
New four-seed extension	42001, 42002, 42003, 42004	Valid	Primary DistDNA evidence

The current 98.2% grid supports

- Two-round cosine, linear SVM, and RBF-SVM comparisons.
- Temperature/top-p robustness analysis on a fixed matched cohort.
- Exploratory representation comparisons when every method reads identical artifacts.

It does not yet support

- A leakage-free same-setting K=2 reference and K=2 query split.
- A claim that seed 42 duplicates or unknown-seed runs are independent samples.
- A fair DistDNA advantage claim based only on the easier single-vector cosine baseline.

Current seed-clean inventory

Across the 12 stochastic settings, the live repository contains **320 unique successful model–setting artifacts** with explicit seed 42001 or 42002. A further 176 successful stochastic artifacts use seed 42 and are not counted as seed-clean four-seed evidence. The 100 deterministic artifacts are also excluded because repeated sampling is inactive at T=0.

Operational rule for the new queue: skip an artifact only when the same model, decoding setting, prompt set, token budget, and exact generation seed already have a successful output. Unknown-seed and seed-42 artifacts remain in the historical grid but do not satisfy a 42001–42004 slot.

3. The immediate DistDNA pilot

The pilot asks one narrow question: when the generated-response budget is matched, does a distributional representation identify the model more reliably than averaging ordinary DNA vectors and using cosine retrieval?

Same-setting K=2 DistDNA pilot: leakage-free, matched-budget comparison

Preselect a difficult 20–30 model cohort; use identical prompts, setting, encoder, and token budget.

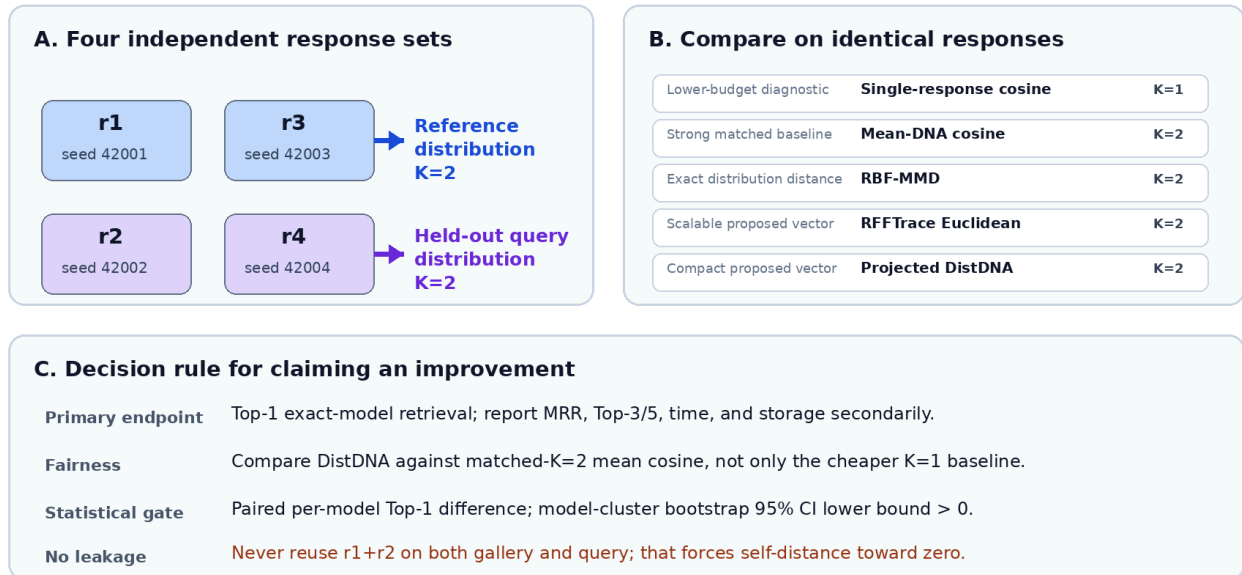


Figure 2. Four independent response sets at one fixed decoding setting. Seeds 42001 and 42003 form the reference distribution; seeds 42002 and 42004 form the held-out query distribution. Every K=2 method reads the same four sets.

Recommended first setting

T=0.7, top-p=1.0. This is deliberately stochastic and therefore more likely to reveal whether distributional information helps. It also has broad explicit seed-42002 coverage in the current grid, reducing the amount of new generation needed for the later 100-model extension.

Why four seeds are needed

Two seeds are used to estimate the reference distribution and two different seeds estimate the query distribution. Reusing a response set on both sides would make a model artificially close to itself. Four seeds are therefore the minimum clean design for a K=2 versus K=2 same-setting comparison.

What the 10-model result means

The 10-model run is a debugging and effect-direction pilot. It can reveal implementation errors, leakage, saturation, or an obviously unhelpful representation. It is not a final statistical claim: with ten queries, one corrected model changes Top-1 by ten percentage points. If the pipeline behaves sensibly, expand to 20–30 deliberately confusable models before interpreting confidence intervals.

4. Proposed 10-model small-scale cohort

All candidates below are already members of the fixed 100-model cohort and are advertised below 0.8B parameters. Selection is based on scale and family structure, not on DistDNA outcomes. The repeated Pythia and Qwen families intentionally make the retrieval task less likely to saturate.

#	Model	Scale	Role in pilot
1	ibm-granite/granite-4.0-h-350m-base	≈0.35B	Granite baseline
2	EleutherAI/pythia-410m-seed1	≈0.41B	Confusable same-family trio
3	EleutherAI/pythia-410m-seed2	≈0.41B	
4	EleutherAI/pythia-410m-seed3	≈0.41B	
5	unsloth/Qwen2.5-0.5B	≈0.50B	Related Qwen variants
6	unsloth/Qwen2.5-0.5B-Instruct	≈0.50B	
7	Ihor/Text2Graph-R1-Qwen2.5-0.5b	≈0.50B	
8	robowaifudev/megatron-gpt2-345m	≈0.35B	GPT-2 architecture
9	state-spaces/mamba-370m-hf	≈0.37B	State-space architecture
10	unsloth/Qwen3-0.6B-Base	≈0.60B	Newer Qwen family

Pre-launch check. Parameter labels above are model-name/model-card scale labels. The runner should still validate configuration compatibility, non-empty generation, tokenizer access, and cached model availability before the four-seed queue starts.

Frozen generation contract

Decoding setting	T=0.7, top-p=1.0, sampling enabled
Seeds	42001, 42002, 42003, 42004
Prompts	The same 100 classical-Chinese prompts already used by the primary grid
Response budget	The same maximum generated-token limit as the primary grid
Representation	The same response encoder and projection configuration for every method
Output policy	Never overwrite historical artifacts; write to a seed-explicit extension directory

5. Fair comparison and expected pilot time

Methods evaluated on identical responses

Method	Sample budget	Purpose	Interpretation
Single-response cosine	K=1 vs K=1	Original low-budget diagnostic	Useful, but not the fairness gate
Mean-DNA cosine	K=2 vs K=2	Strong matched-budget baseline	Primary comparator
Exact RBF-MMD	K=2 vs K=2	Exact distribution distance	Accuracy reference for approximation
RFFTrace	K=2 vs K=2	Scalable distributional vector	Primary DistDNA result
Compact DistDNA	K=2 vs K=2	Projected representation	Accuracy-storage trade-off

The decision rule is simple: for each held-out query distribution, rank the ten reference models by distance and check whether the true model is ranked first. Report Top-1, MRR, Top-3, runtime, and storage. The meaningful improvement is **DistDNA minus mean-DNA cosine** on the same query models.

Time from an available four-GPU A100 allocation

0-0.5 h	Audit reusable seed-explicit artifacts, freeze the ten-model manifest, and dry-run the queue.
0.5-3.5 h	Generate the missing seed-specific responses. Historical small-model observations have a 2.4-minute median per artifact, with a long retry tail; 1–3 hours is the realistic wall-clock range on four GPUs.
3.5-4.5 h	Build embeddings and run cosine, exact MMD, RFFTrace, and compact DistDNA on the identical split.
4.5-8 h	Inspect nearest-neighbour errors, verify no leakage, adjust implementation if necessary, rerun evaluation, and produce the first result table.

1-3 h expected generation time for the 10-model pilot	4-8 h expected time to a checked first DistDNA comparison
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Practical advantage. Changes to RFF dimension, bandwidth, projection, or the DistDNA vector do not require regenerating model responses. Once the four raw response sets are stored correctly, representation development can iterate quickly.

6. Scaling from 10 models to the full cohort

“Four seeds for 100 models” has two materially different interpretations. The operational plan should name the scope explicitly.

Scope	Target artifacts	Reusable now	Still required	Expected wall time
One setting T=0.7, top-p=1.0	$100 \times 4 = 400$	≈103 explicit 42001/42002 artifacts	≈297	4 A100s: 2–4 days 8 A100s: 1.5–3 days
All 12 stochastic settings	$100 \times 12 \times 4 = 4,800$	320 explicit 42001/42002 artifacts	4,480	4 A100s: 4–6 weeks 8 A100s: 2–3 weeks

Theoretical compute for the all-setting extension is about 2,335 A100 GPU-hours, estimated from the 2,552 successful primary artifacts (31.3 GPU-minutes per artifact on average). Wall-clock ranges add allowance for retries, timeouts, incompatible models, scheduling gaps, and audit passes.

Recommended staged execution

Stage A	Let the current 72-slot recovery pass finish and freeze its final status. This preserves a clean endpoint for the original two-round primary grid.
Stage B	Run the 10-model four-seed pilot at T=0.7, top-p=1.0. Obtain a checked DistDNA versus matched mean-cosine result within one working day.
Stage C	Start the 100-model extension at the same setting. Skip only successful artifacts with the exact required seed; regenerate unknown-seed and seed-42 slots.
Stage D	While Stage C runs, iterate on DistDNA using already generated raw responses. No new generation is needed for representation-only changes.
Stage E	Expand to all settings only if the pipeline passes leakage checks and DistDNA shows a useful effect or a clearly motivated diagnostic value.

Why not immediately queue all 4,480 missing artifacts? It is safe to generate raw responses while representation code changes, but the full queue commits several weeks of A100 time. A same-day pilot can first catch split errors, saturated retrieval, an unsuitable bandwidth, or a DistDNA construction that needs revision.

Seed replacement policy

Existing unknown-seed and seed-42 artifacts are retained for provenance and the original analysis. They are not deleted or overwritten. The four-seed extension writes new, seed-explicit artifacts alongside them and the audit chooses the correct artifact by exact seed.

7. Implementation status and next deliverables

Status	Component	Repository location
Implemented	Exact MMD, shared RFF map, RFFTrace, compact projection	src/llm_dna/distributional.py
Implemented	Matched-sample retrieval evaluator and export	scripts/run_rfftrace_experiment.py
Implemented	Optional DistDNA multiclass SVM evaluation	scripts/evaluate_rfftrace_svm.py
Verified	Distributional and evaluator unit tests	tests/test_distributional.py tests/test_rfftrace_experiment.py
Next	Seed-exact audit and four-seed generation queue	New pilot manifest and resumable runner
Next	Pilot result table, paired errors, and saturation diagnostic	Generated after the four-seed pilot

Acceptance checks before interpreting accuracy

1. **Exact split:** no response artifact appears in both reference and query distributions.
2. **Matched budget:** DistDNA and mean-DNA cosine use the same four response sets.
3. **Shared map:** every model uses the same response encoder, RFF parameters, and optional projection.
4. **No outcome-based cohort changes:** the ten models are frozen before DistDNA accuracy is inspected.
5. **Failure transparency:** failed/empty responses are reported and never silently replaced by a different model.
6. **Saturation check:** if all methods score 100%, move to a larger or more confusable cohort rather than claiming equivalence or improvement.

Deliverable sequence

- 1 Final recovery audit
- 2 10-model seed manifest
- 3 Four-seed generation
- 4 Matched comparison
- 5 Error analysis
- 6 100-model scale-out

Bottom line

A first technically meaningful DistDNA comparison is a same-day task once GPUs are available. A full 100-model, one-setting four-seed dataset is a multi-day task. A four-seed replacement across the entire 12-setting grid is a separate multi-week compute commitment and should follow, not precede, the pilot gate.

Provenance. Counts in this note were generated from the live repository on 13 August 2026 at 13:56 SGT. Recovery was still running. Final figures must be regenerated after that process exits. The 5 August report and PDF remain unchanged.